

F24-016-D-InsightWire

Project Team

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Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	2
1.3	Modules	2
1.3.1	Module 1: News Aggregation and Personalization	2
1.3.2	Module 2: Chatbot and User Interaction	2
1.3.3	Module 3: Bias Detection and Story Comparison	3
1.3.4	Module 4: Contextual Background and Analytics	3
1.4	User Classes and Characteristics	3
2	Project Requirements	5
2.1	Functional Requirements	5
2.1.1	Module 1: News Aggregation, Categorization, and Personalization	5
2.1.2	Module 2: Chatbot, User Interaction, and Feedback	6
2.1.3	Module 3: Bias Detection and Story Comparison	6
2.1.4	Module 4: Contextual Background and News Analytics	6
2.2	Non-Functional Requirements	7
2.2.1	Reliability	7
2.2.2	Usability	7
2.2.3	Performance	7
2.2.4	Scalability	7
2.2.5	Portability	8
2.2.6	Security	8
2.2.7	Availability	8
2.3	Domain Model	8
2.4	Use-case/Event Response Table/Storyboarding	9
2.4.1	Use Case Diagram	9
2.4.2	Extended Usecases	11
2.4.2.1	Use Case 1	11
2.4.2.2	Use Case 2	12
2.4.2.3	Use Case 3	13
2.4.2.4	Use Case 4	14

2.4.2.5	Use Case 5	15
2.4.2.6	Use Case 6	16
2.4.2.7	Use Case 7	17
2.4.2.8	Use Case 8	18
2.4.2.9	Use Case 9	19
2.4.2.10	Use Case 10	20
2.4.2.11	Use Case 11	21
2.4.2.12	Use Case 12	22
3	System Overview	25
3.1	Architectural Design	25
3.2	Design Models	25
3.2.1	Activity Diagrams	25
3.2.2	Class Diagram	31
3.2.3	Class-Level Sequence Diagrams	32
3.3	Data Design	38
	References	41

List of Figures

2.1	Domain Model	9
2.2	Use Case Diagram	10
3.1	Architecture Diagram	25
3.2	Aggregation and Personalization	26
3.3	Bias Detection	27
3.4	Context Background	28
3.5	LLM and RAG Chatbot	29
3.6	News Analytics	30
3.7	Story Comparison	31
3.8	Class Diagram	32
3.9	Sequence Diagram for Aggregation and Personalization	33
3.10	Sequence Diagram for Bias Analysis	34
3.11	Sequence Diagram for Story Comparison	35
3.12	Sequence Diagram for Contextual Background	36
3.13	Sequence Diagram for LLM and RAG Chatbot	37
3.14	Sequence Diagram for News Analytics	38
3.15	data Design Model	39

List of Tables

2.1	Use Case Table for Summary Generation	12
2.2	Use Case Table for Interaction with AI Chatbot	13
2.3	Use Case Table for Signup	14
2.4	Use Case Table for Customize News Feeds	15
2.5	Use Case Table for Read Articles	15
2.6	Use Case Table for View Summary	16
2.7	Use Case Table for Compare Articles	18
2.8	Use Case Table for Submit Feedback	19
2.9	Use Case Table for Login	20
2.10	Use Case Table for Bias Detection	21
2.11	Use Case Table for Manage User Profile	22
2.12	Use Case Table for Provide News	23

Chapter 1

Introduction

This project proposal introduces InsightWire, an innovative and comprehensive news platform designed to transform how users consume and understand news. InsightWire addresses the limitations of traditional news consumption by integrating features for news aggregation, bias detection, and contextual analysis. Traditional news sources often present information with bias and limited context, making it difficult for users to grasp the full picture. Our project aims to overcome these challenges by providing a comprehensive solution that aggregates stories from multiple sources at one place, tracks how narratives evolve over time, and checks for political biases. The platform has a bias detection system, a timeline feature that brings up the contextual background and news analytics to offer users a deeper understanding of the news landscape. Our project's combination of these features offers a more informed and nuanced perspective on current events, giving our users the ability to stay informed and think critically about the information they consume. [1].

1.1 Problem Statement

In today's Information Age, there is an abundance of news media. This has many benefits but it comes with its down sides too. One of the major ones that has become very obvious these days is the polarization caused in society by biased news sources. These news companies are often run by giant corporations that have their own agendas and want to control the narrative to further their own political and social plans. The lay man blindly trusts these news sources.

1.2 Scope

The scope of the InsightWire project is to provide a news platform that addresses the issues of bias, lack of context, and fragmented information in news consumption. The system is designed to aggregate news articles from 18 to 20 sources to present a balanced view of current events. We are only concerned with political news, nonpolitical news is outside our scope. It includes a bias detection mechanism that classifies news articles into various political leanings, helping users identify and understand inherent biases. InsightWire also features a contextual background component that creates a timeline of events as they have unfolded and led up to the present story. Additionally, the platform has news analytics to identify and assess any forward looking predictions or trends mentioned in the articles. However, the project does not extend to predictive analytics based on historical data or in-depth investigative journalism. It focuses solely on the aggregation, bias detection, contextual analysis, and trend evaluation of news articles as presented in the platform. The scope also excludes real-time fact-checking and user-generated content moderation, which are outside the boundaries of this project.

1.3 Modules

1.3.1 Module 1: News Aggregation and Personalization

This module focuses on collecting news from diverse sources and categorizing them for easy consumption. It also offers personalization features to tailor news feeds based on user preferences and highlights local news. The goal is to provide users with a balanced and customized news experience.

1. News Aggregation
2. Categorization and Tagging
3. Customization and Personalization
4. Local News Highlighting

1.3.2 Module 2: Chatbot and User Interaction

This module includes a chatbot that interacts with users, summarizing articles and answering news-related queries. Additionally, it integrates user feedback features where users can rate and comment on articles, enhancing overall engagement and content quality.

1. Chatbot
2. LLM and RAG Integration
3. Users Feedback

1.3.3 Module 3: Bias Detection and Story Comparison

This module is dedicated to analyzing the political bias of news sources and allowing users to compare how different outlets report on the same story. It also includes a voiceover feature to improve accessibility and engagement, especially for users on the go.

1. Bias Detection
2. Story Comparison
3. Voice Over

1.3.4 Module 4: Contextual Background and Analytics

This module provides a deeper understanding of news articles by offering contextual background, linking stories to historical events, and providing analytics to help users anticipate potential future developments. It enhances the user's grasp of the broader context behind the news.

1. Contextual Background
2. News Analytics

1.4 User Classes and Characteristics

User class	Description
General News Consumers	This class represent every user who seek a reliable and balance source of News. They may use InsightWire to access News from multiple sources, Customize feed according to their personal preferences, view balance news and stay informed about current event.
Researchers and Analysts	This class includes professional, academics and journalists who use the platform for the in-depth research on news trends, bias patterns, and story evolution over the time. They rely on insightWire's analytics and contextual data to conduct comprehensive analysis.
News Providers	News agencies, journalists, or media outlets that contribute content to the insightWire may use platform to reach the broader users.
IT and Development	This user represent the internal team responsible for development, maintenance and updating the system. They focus on technical development, back-end development and integrating new features to ensure platform scalability and reliability.
Potential Investors	Investors and stockholders who are interested in the growth and potential of the insightWire as a business.

Example: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Functional Requirements

This section describes the functional requirements of the system.

2.1.1 Module 1: News Aggregation, Categorization, and Personalization

Following are the functional requirements for Module 1:

1. **FR1:** The system shall retrieve and aggregate news articles from a minimum of 18 sources and a maximum of 20 sources.
2. **FR2:** The system shall only retrieve news articles that are categorized as political; non-political articles shall be excluded during the aggregation process.
3. **FR3:** The system shall group each retrieved article by subject matter, ensuring that articles about the same event are grouped together.
4. **FR4:** The system shall tag each news article with publication source, publication date, and political bias classification.
5. **FR5:** The system shall allow users to create their profiles.
6. **FR6:** The system shall enable users to customize their news feed by selecting specific interests and desired levels of political balance.

7. **FR7:** The system shall determine a user's location to highlight news articles relevant to that location.

2.1.2 Module 2: Chatbot, User Interaction, and Feedback

Following are the functional requirements for Module 2:

1. **FR1:** The system shall include a chatbot that can respond to user queries.
2. **FR2:** The chatbot shall be able to summarize the contents of specific news articles.
3. **FR3:** The chatbot shall provide insights into themes or trends across a minimum of three sources for the same news story.
4. **FR4:** The system shall provide users the ability to submit reviews on news articles.
5. **FR5:** The system shall allow users to leave comments on news articles, which will be stored and displayed alongside the article.

2.1.3 Module 3: Bias Detection and Story Comparison

Following are the functional requirements for Module 3:

1. **FR1:** The system shall analyze the political bias of each news article, classifying it into one of the following categories: left, center, right.
2. **FR2:** The system shall, upon user request, display articles from two sources on the same topic, with differences in language and tone highlighted for comparison.
3. **FR3:** The system shall generate a summary of the detected bias for each news story, making the overview accessible to the user with a single click.

2.1.4 Module 4: Contextual Background and News Analytics

Following are the functional requirements for Module 4:

1. **FR1:** The system shall display a timeline of related historical events for each news story, allowing users to view how the story has evolved over time.
2. **FR2:** The system shall perform analytics on the news articles, identifying trends or predictions explicitly mentioned within the content.
3. **FR3:** The system shall allow users to filter the contextual timeline by date, displaying events that occurred within a specified timeframe.

2.2 Non-Functional Requirements

This section specifies the non-functional requirements. These quality requirements are specific, quantitative, and verifiable, ensuring the system functions as intended beyond its core features.

2.2.1 Reliability

1. **RL1:** The system shall provide 99.9% uptime during normal operations to ensure continuous availability for users.
2. **RL2:** The system shall ensure that the aggregation of news sources continues even if up to 2 external news sources fail to deliver data.

2.2.2 Usability

1. **US1:** The system shall provide an intuitive user interface with clear navigation paths.
2. **US2:** The system shall allow users to customize the interface to enhance the user experience.

2.2.3 Performance

1. **PE1:** The system shall handle up to 200 concurrent users without noticeable degradation in performance.
2. **PE2:** The system shall ensure that the response time of the chatbot remains under 10 seconds.
3. **PE3:** The system shall support the retrieval of news analytics in under 5 minutes.

2.2.4 Scalability

1. **SC1:** The system shall store news articles and user data in a distributed NoSQL database (e.g., MongoDB) that allows for scaling data storage as the number of users and articles increases.

2.2.5 Portability

1. **PR1:** The system shall be accessible on Chrome without requiring additional plug-ins or extensions.
2. **PR2:** The system shall provide a responsive design that works seamlessly across desktop, tablet, and mobile devices with varying screen sizes.

2.2.6 Security

1. **SE1:** The system shall require all user authentication to be conducted via HTTPS, ensuring that all data transmitted between the client and server is encrypted.
2. **SE2:** The system shall implement role-based access control (RBAC) to restrict access to sensitive operations, such as content moderation and system configuration, to authorized personnel only.
3. **SE3:** The system shall enforce strong password requirements (minimum 8 characters, including at least one uppercase letter, one number, and one special character) for user accounts.
4. **SE4:** The system shall prevent cross-site scripting (XSS) attacks by sanitizing user inputs.

2.2.7 Availability

1. **AV1:** The system shall be available 24/7, with planned maintenance periods not exceeding 2 hours per month.
2. **AV2:** The system shall notify users of any planned maintenance or outages at least 24 hours in advance, displaying alerts in the application and via email.

2.3 Domain Model

Domain Model of the project is given below

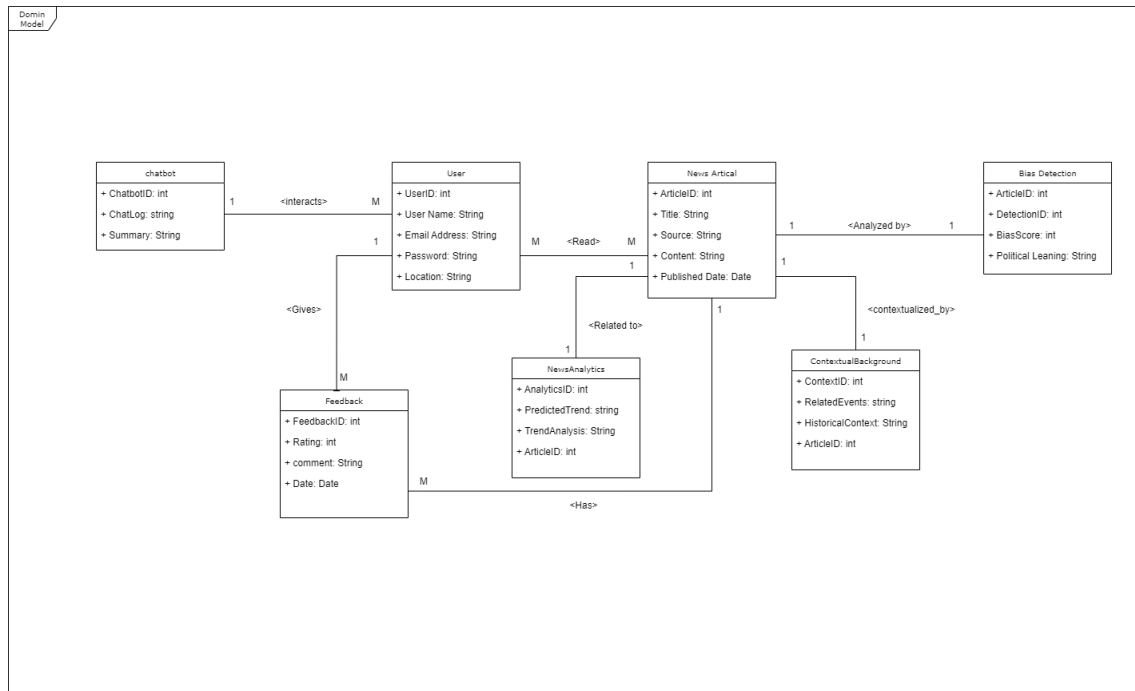
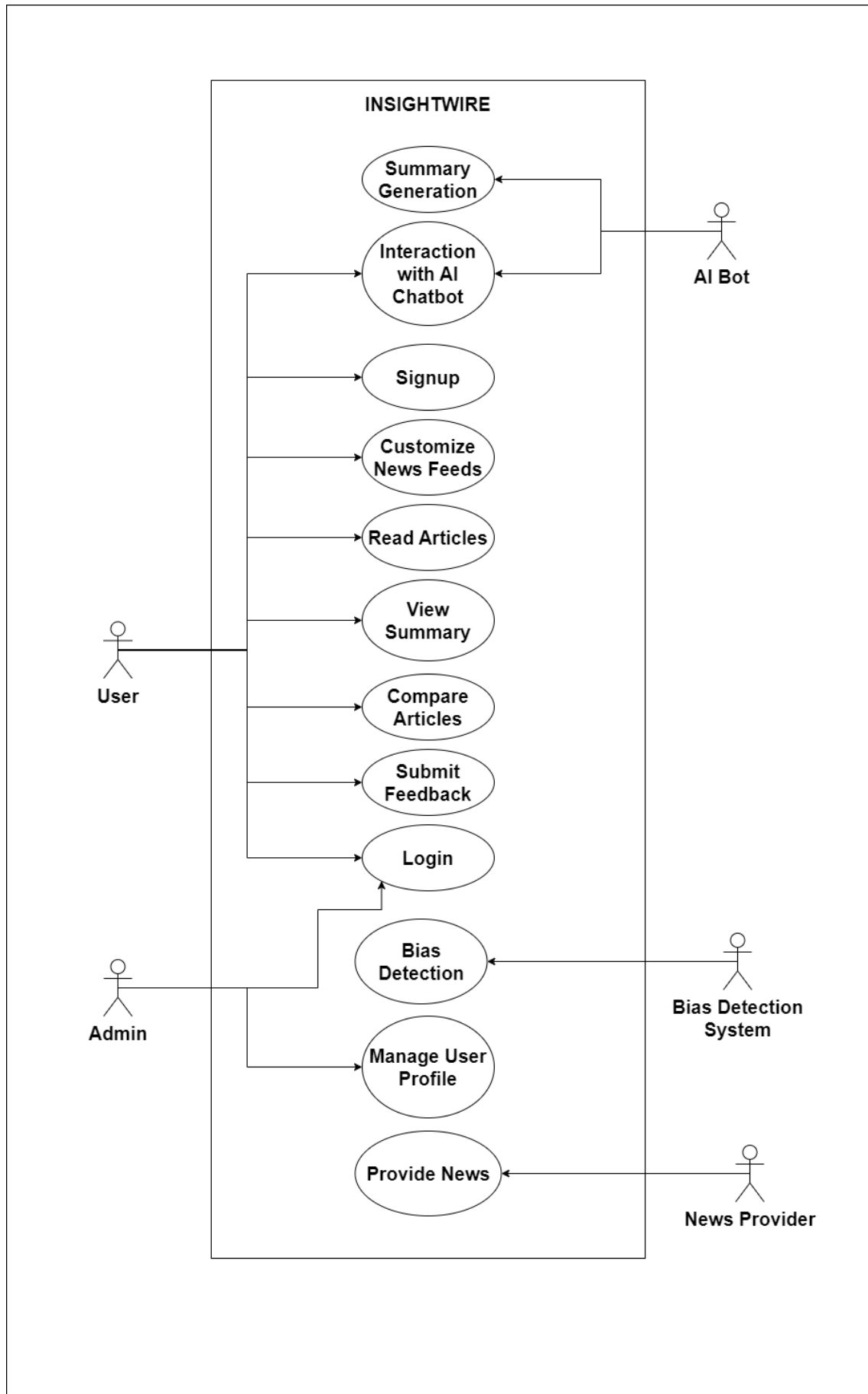


Figure 2.1: Domain Model

2.4 Use-case/Event Response Table/Storyboarding

2.4.1 Use Case Diagram

Use case diagram of the project is given below



10
Figure 2.2: Use Case Diagram

2.4.2 Extended Usecases

2.4.2.1 Use Case 1

Use Case No.	UC-01
Name	Summary Generation
Description	The system generates summaries of articles that have undergone bias detection.
Priority	High
Actors	AI Bot
Preconditions	Articles must be scraped, categorized, and analyzed for bias.
Main Scenario	1. The AI bot receives the articles after bias detection. 2. The system summarizes the key points of the articles. 3. Summaries are stored in the database and linked to the original articles.
Postcondition	The summary is successfully generated and linked with the article for user access.
Exception	If the article is too short or lacks sufficient content, a summary may not be generated.

Table 2.1: Use Case Table for Summary Generation

2.4.2.2 Use Case 2

Use Case No.	UC-02
Name	Interaction with AI Chatbot
Description	Users can interact with the AI chatbot to ask questions and get information based on the articles.
Priority	High
Actors	User, AI Bot
Preconditions	Users must be logged in to access the chatbot.
Main Scenario	1. The user initiates a conversation with the AI chatbot. 2. The user asks a question related to the articles. 3. The AI bot processes the request and provides a relevant response.
Postcondition	The user receives an answer to their question.
Exception	If the AI bot cannot understand the question, it prompts the user to rephrase.

Table 2.2: Use Case Table for Interaction with AI Chatbot

2.4.2.3 Use Case 3

Use Case No.	UC-03
Name	Signup
Description	New users can create an account on the platform.
Priority	High
Actors	User
Preconditions	None
Main Scenario	1. The user accesses the signup page. 2. The user fills in the required information (name, email, password). 3. The user submits the signup form. 4. The system creates a new user account and sends a confirmation email.
Postcondition	The user account is created, and the user can log in.
Exception	If the email is already in use, the user is notified to choose another email.

Table 2.3: Use Case Table for Signup

2.4.2.4 Use Case 4

Use Case No.	UC-04
Name	Customize News Feeds
Description	Users can personalize their news feed based on their preferences.
Priority	Medium
Actors	User
Preconditions	The user must be logged in.
Main Scenario	1. The user accesses the settings page. 2. The user selects categories or topics of interest. 3. The user saves the changes. 4. The system updates the news feed based on user preferences.
Postcondition	The news feed reflects the user's selected preferences.
Exception	If no preferences are selected, the default feed is displayed.

Table 2.4: Use Case Table for Customize News Feeds

2.4.2.5 Use Case 5

Use Case No.	UC-05
Name	Read Articles
Description	Users can read full articles from the news feed.
Priority	High
Actors	User
Preconditions	The user must be logged in.
Main Scenario	1. The user selects an article from the news feed. 2. The system displays the full article content. 3. The user reads the article.
Postcondition	The user has read the article.
Exception	If the article is not available, a message is displayed.

Table 2.5: Use Case Table for Read Articles

2.4.2.6 Use Case 6

Use Case No.	UC-06
Name	View Summary
Description	Users can view the generated summary of articles.
Priority	High
Actors	User
Preconditions	The articles must have summaries generated.
Main Scenario	1. The user selects an article from the news feed. 2. The system displays the article summary. 3. The user reads the summary.
Postcondition	The user has viewed the article summary.
Exception	If no summary is available, a message is displayed.

Table 2.6: Use Case Table for View Summary

2.4.2.7 Use Case 7

Use Case No.	UC-07
Name	Compare Articles
Description	Users can compare two or more articles based on selected criteria.
Priority	Medium
Actors	User
Preconditions	The user must select articles to compare.
Main Scenario	1. The user selects multiple articles from the news feed. 2. The user chooses comparison criteria (e.g., bias, source). 3. The system displays a side-by-side comparison of the articles.
Postcondition	The user can analyze the differences between the articles.
Exception	If insufficient articles are selected, a message is displayed.

Table 2.7: Use Case Table for Compare Articles

2.4.2.8 Use Case 8

Use Case No.	UC-08
Name	Submit Feedback
Description	Users can provide feedback on articles and the overall platform.
Priority	Medium
Actors	User
Preconditions	The user must be logged in.
Main Scenario	1. The user navigates to the feedback section. 2. The user fills in the feedback form. 3. The user submits the feedback. 4. The system acknowledges the receipt of feedback.
Postcondition	The feedback is stored for review.
Exception	If the feedback form is incomplete, a message prompts the user to complete it.

Table 2.8: Use Case Table for Submit Feedback

2.4.2.9 Use Case 9

Use Case No.	UC-09
Name	Login
Description	Users can log into their accounts to access the platform.
Priority	High
Actors	User
Preconditions	The user must have a registered account.
Main Scenario	1. The user navigates to the login page. 2. The user enters their email and password. 3. The user submits the login form. 4. The system validates the credentials and grants access.
Postcondition	The user is logged into the platform.
Exception	If the credentials are incorrect, a message prompts the user to try again.

Table 2.9: Use Case Table for Login

2.4.2.10 Use Case 10

Use Case No.	UC-10
Name	Bias Detection
Description	The system detects and classifies the bias present in articles.
Priority	High
Actors	Bias Detection System
Preconditions	Articles must be scraped and ready for analysis.
Main Scenario	1. The system analyzes the articles for bias. 2. The system classifies the bias into categories (left-wing, right-wing, etc.). 3. The results are stored in the database linked to the articles.
Postcondition	The articles are classified with their respective bias information.
Exception	If the analysis fails, a notification is sent to the admin for manual review.

Table 2.10: Use Case Table for Bias Detection

2.4.2.11 Use Case 11

Use Case No.	UC-11
Name	Manage User Profile
Description	Users can manage their profile information and settings.
Priority	Medium
Actors	User
Preconditions	The user must be logged in.
Main Scenario	1. The user accesses the profile settings page. 2. The user updates their information (name, email, password). 3. The user saves the changes. 4. The system updates the user profile.
Postcondition	The user profile is updated successfully.
Exception	If the information is invalid, an error message prompts for correction.

Table 2.11: Use Case Table for Manage User Profile

2.4.2.12 Use Case 12

Use Case No.	UC-12
Name	Provide News
Description	The system provides users with access to the latest news articles.
Priority	High
Actors	User, Admin
Preconditions	Articles must be scraped and stored in the database.
Main Scenario	1. The user navigates to the news section. 2. The system retrieves the latest articles from the database. 3. The user can browse the articles displayed on the news feed.
Postcondition	The user has access to the latest news articles.
Exception	If no articles are available, a message is displayed indicating that there are no articles to show.

Table 2.12: Use Case Table for Provide News

Chapter 3

System Overview

3.1 Architectural Design

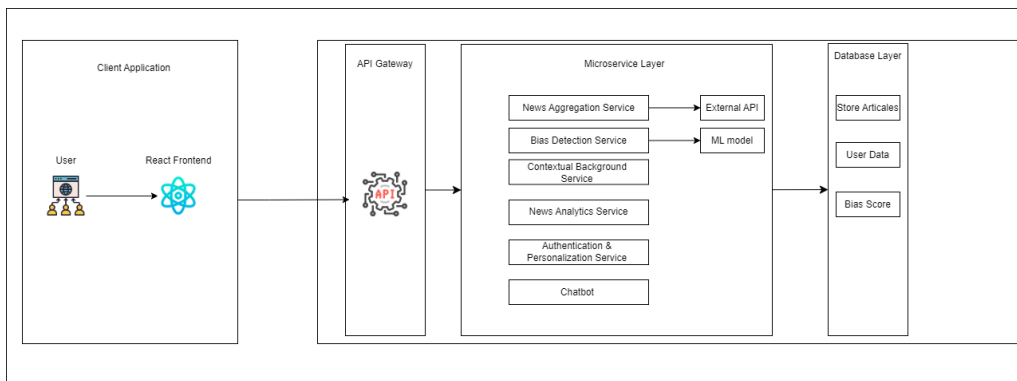


Figure 3.1: Architecture Diagram

3.2 Design Models

3.2.1 Activity Diagrams

1. Activity Diagram 1: Aggregation and Personalization

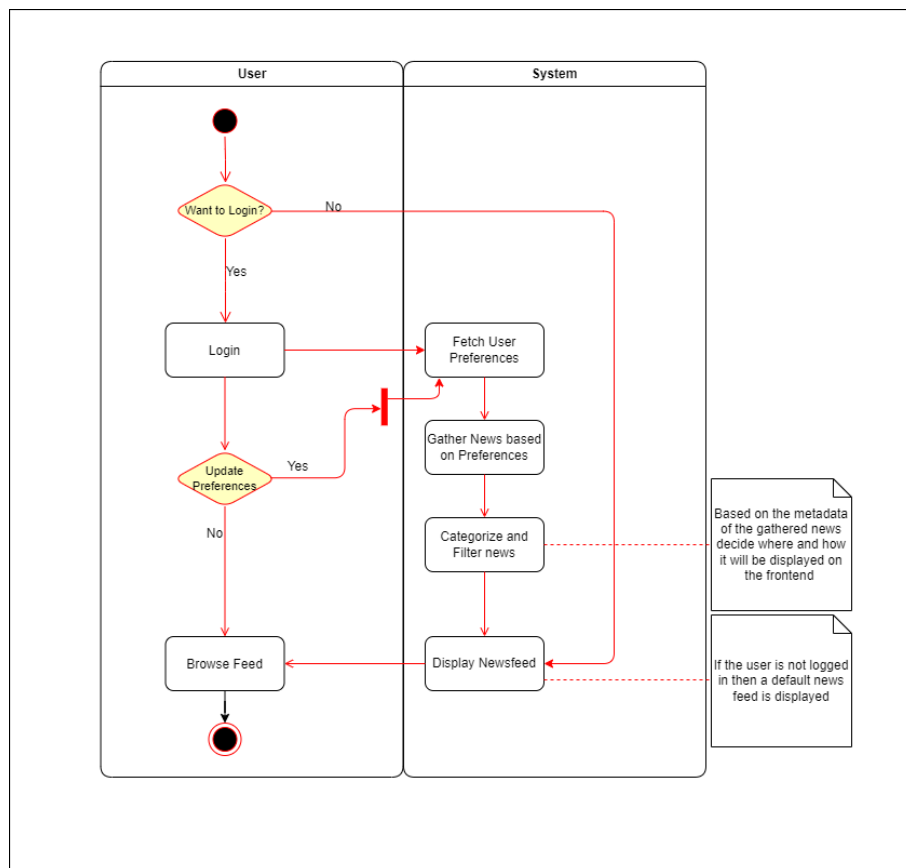


Figure 3.2: Aggregation and Personalization

2. Activity Diagram 2: Bias Detection

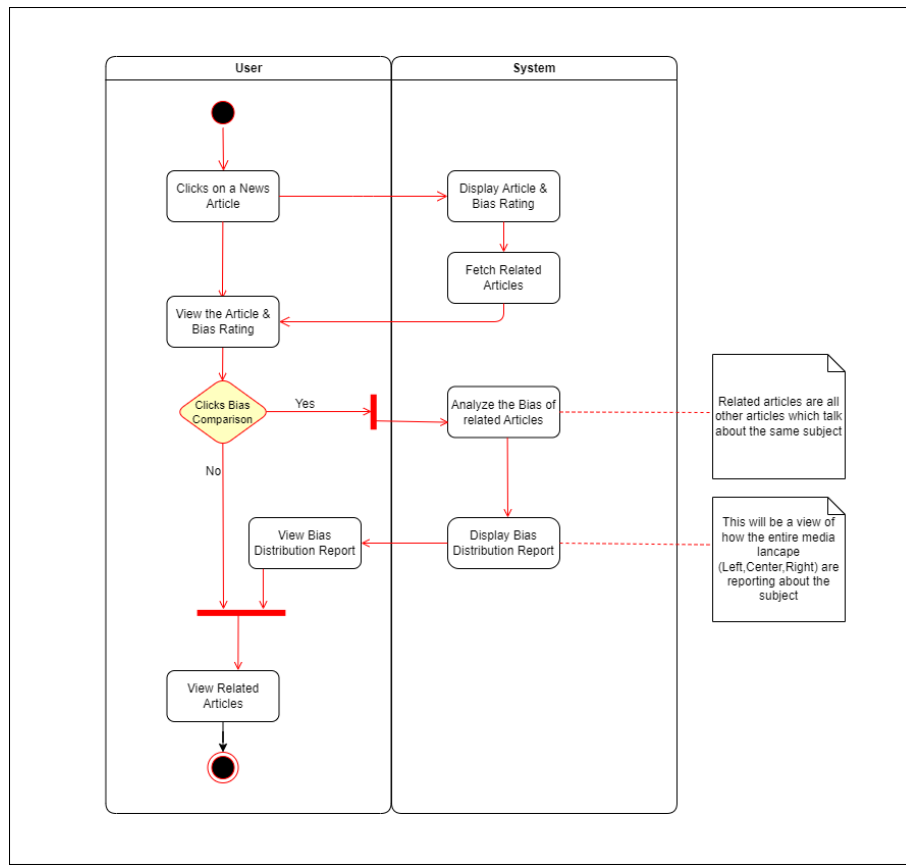


Figure 3.3: Bias Detection

3. Activity Diagram 3: Context Background

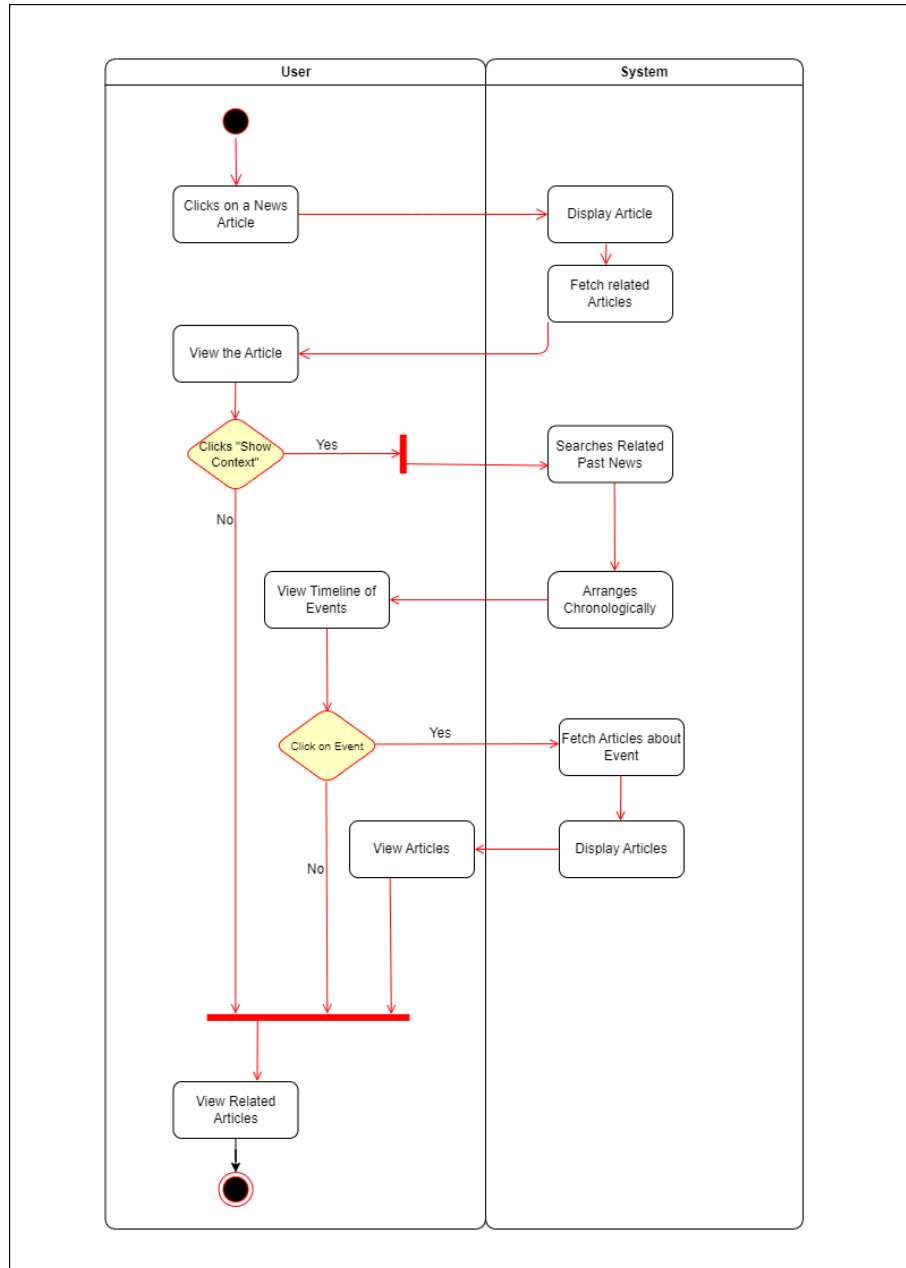


Figure 3.4: Context Background

4. Activity Diagram 4: LLM and RAG Chatbot

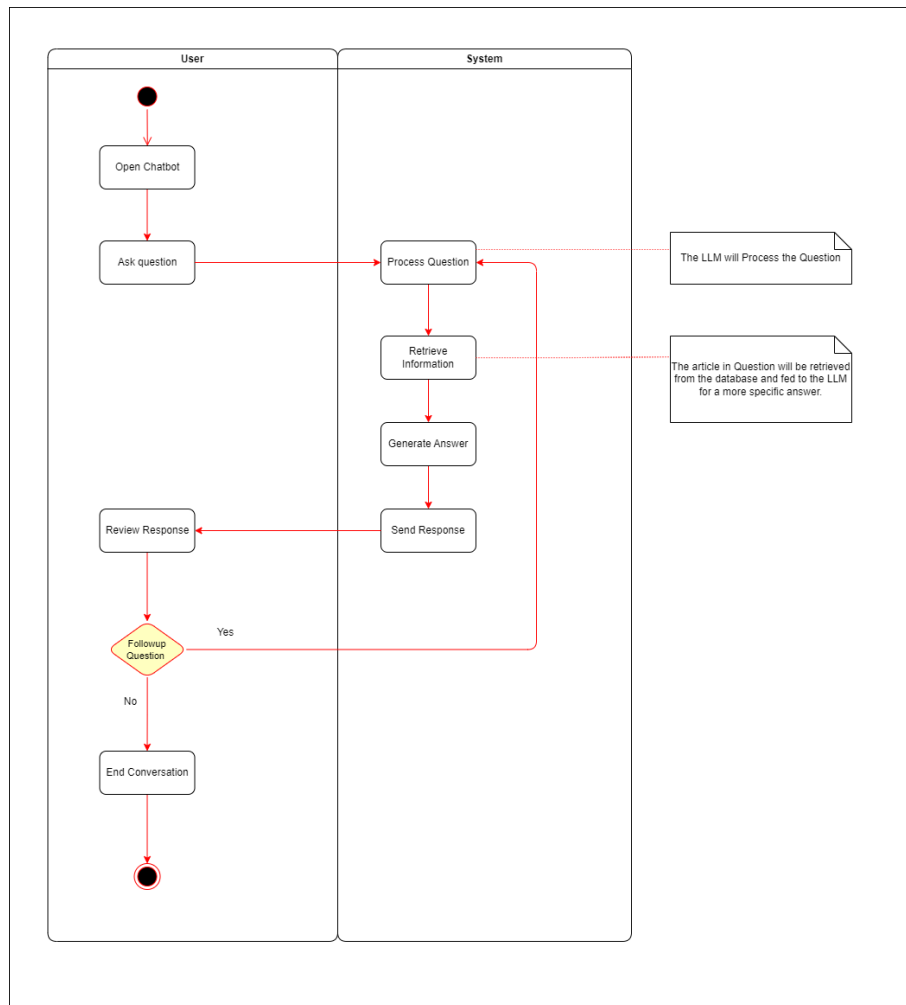


Figure 3.5: LLM and RAG Chatbot

5. Activity Diagram 5: News Analytics

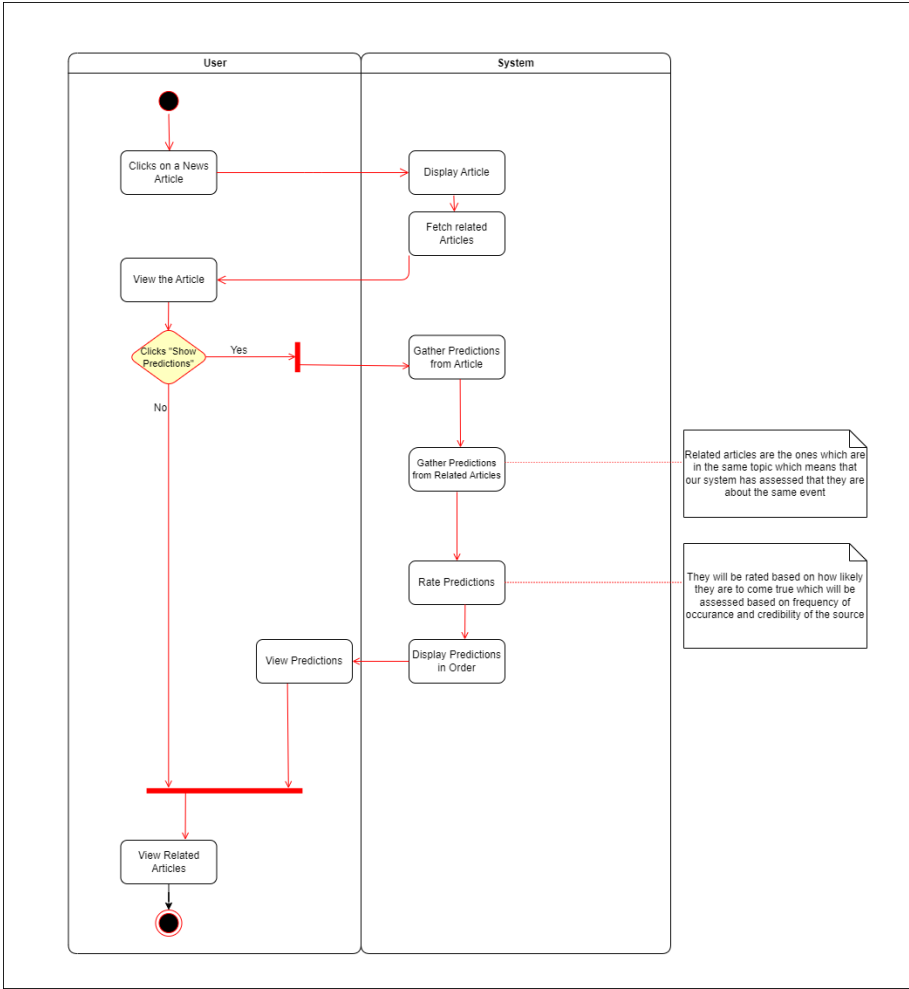


Figure 3.6: News Analytics

6. Activity Diagram 6: Story Comparison

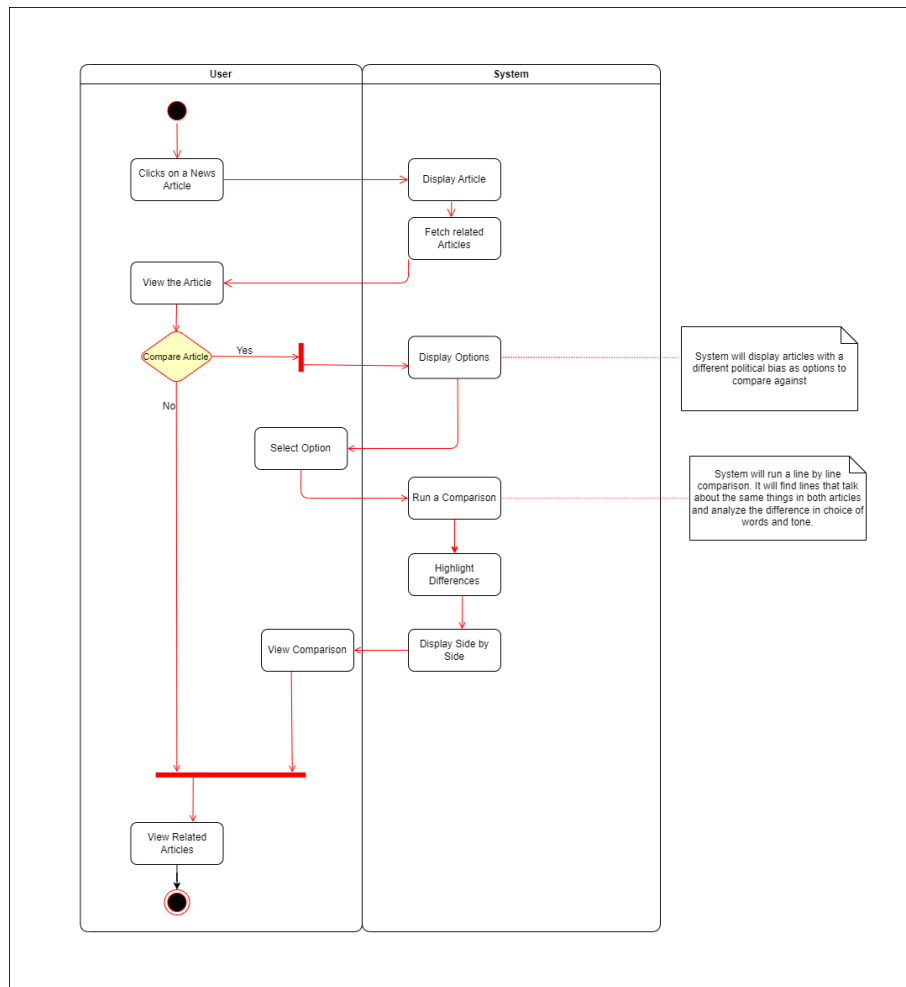


Figure 3.7: Story Comparison

3.2.2 Class Diagram

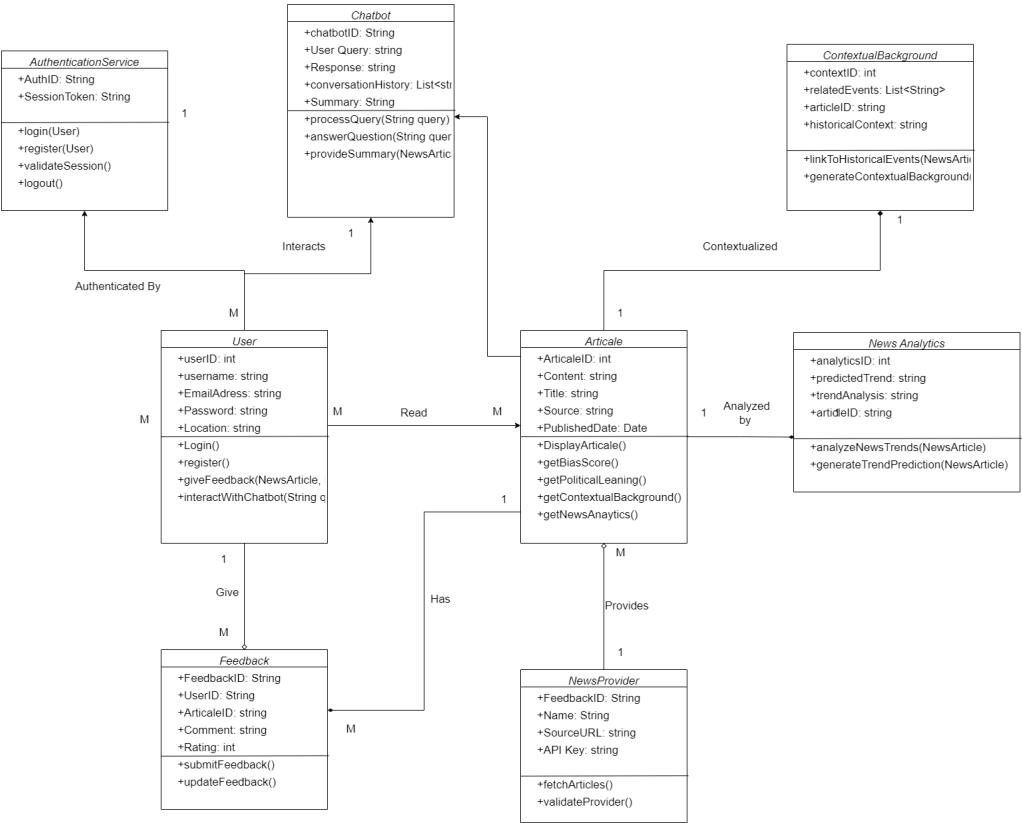


Figure 3.8: Class Diagram

3.2.3 Class-Level Sequence Diagrams

1. Sequence Diagram 1: Aggregation and Personalization

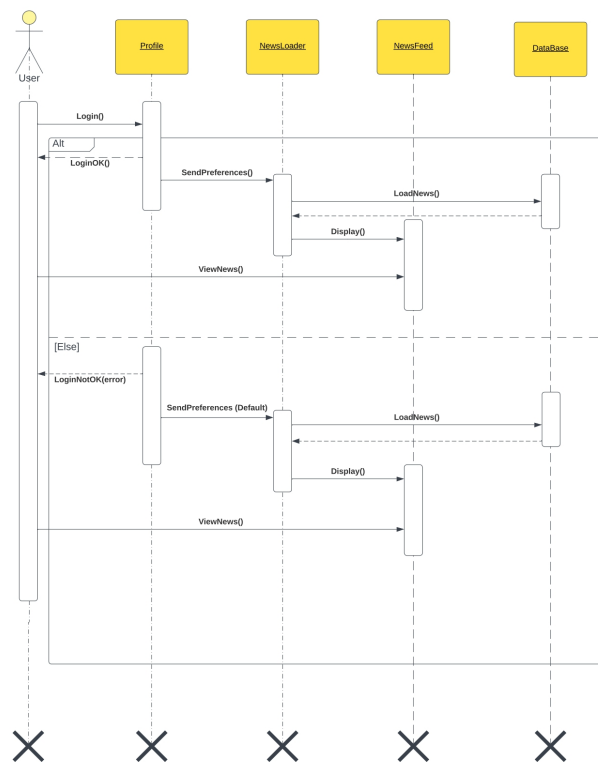


Figure 3.9: Sequence Diagram for Aggregation and Personalization

2. Sequence Diagram 2: Bias Analysis

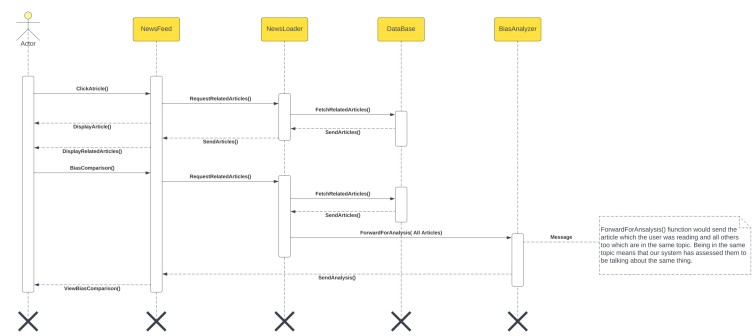


Figure 3.10: Sequence Diagram for Bias Analysis

3. Sequence Diagram 3: Story Comparison

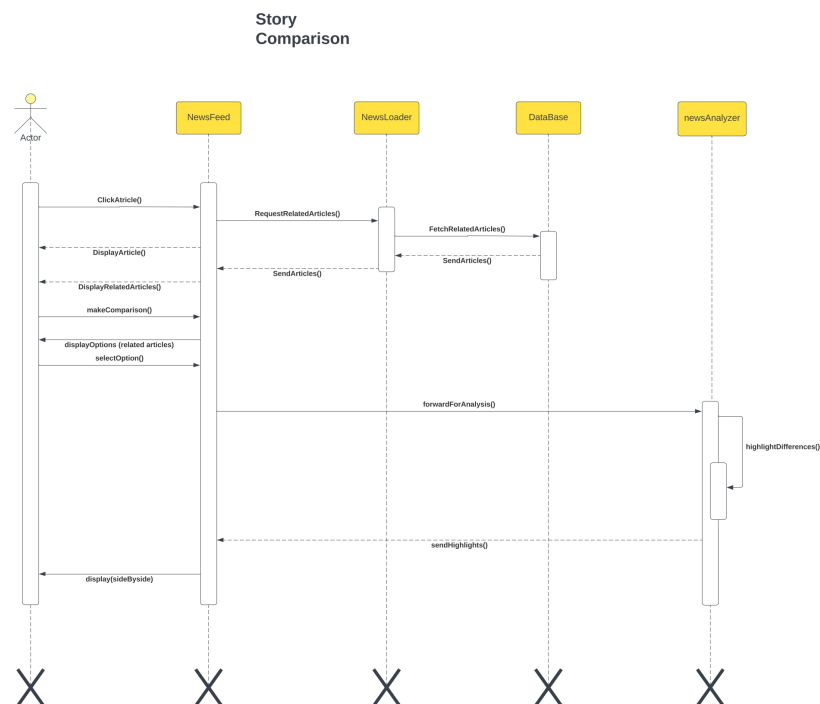


Figure 3.11: Sequence Diagram for Story Comparison

4. Sequence Diagram 4: Contextual Background

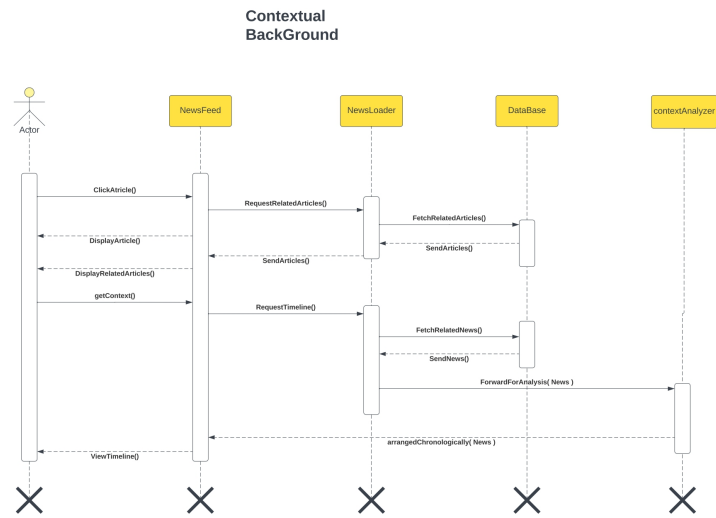


Figure 3.12: Sequence Diagram for Contextual Background

5. Sequence Diagram 5: LLM and RAG Chatbot

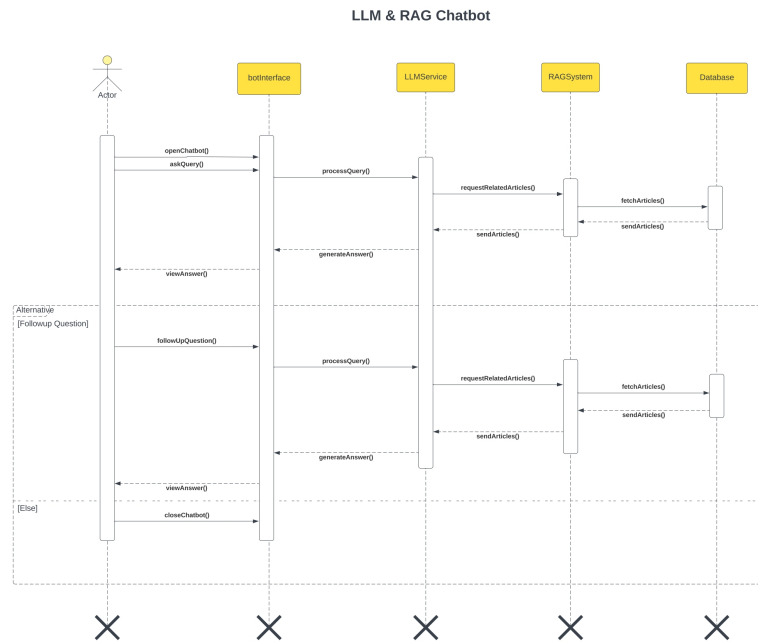


Figure 3.13: Sequence Diagram for LLM and RAG Chatbot

6. Sequence Diagram 6: News Analytics

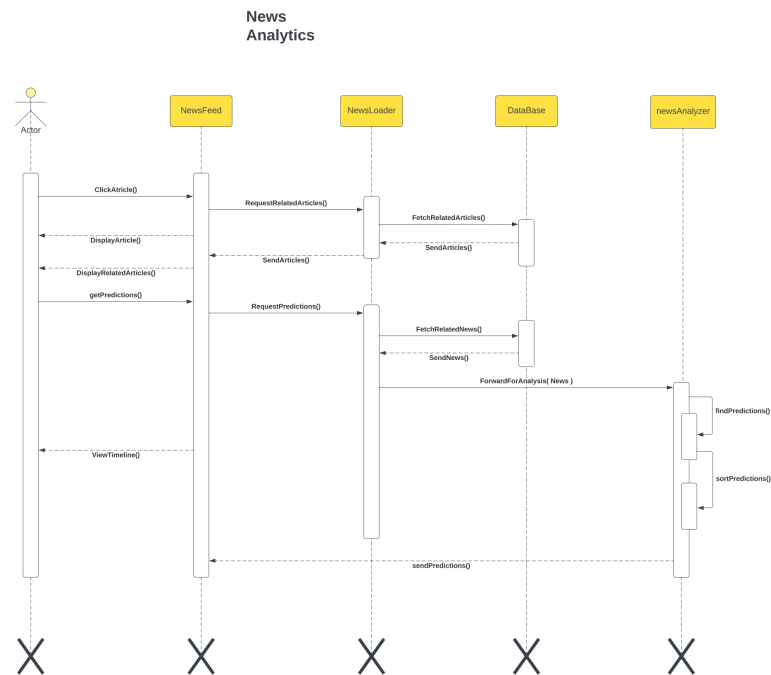


Figure 3.14: Sequence Diagram for News Analytics

3.3 Data Design

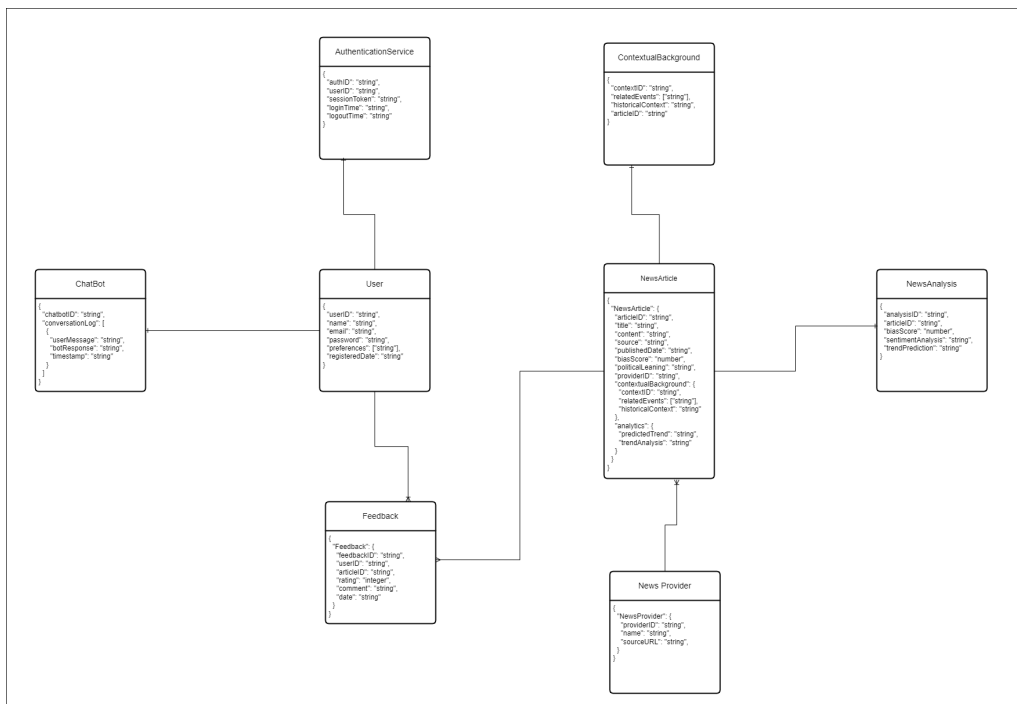


Figure 3.15: data Design Model

Bibliography

- [1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.